

Sai Vineel Reddy Marreddy

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EDUCATION

North Carolina State University, Raleigh, NC, USA

Aug 2024 - Apr 2026

Master of Science in Computer Science

GPA: 3.85/4.0

Coursework: Data Structures, Algorithms, Object Oriented Design, Artificial Intelligence, Machine Learning

Manipal Institute of Technology, Manipal

Jul 2019 - Aug 2023

Bachelor of Technology in Computer Science and Engineering

GPA: 8.35/10

SKILLS

Languages:	C++, Scala, C, Python, JavaScript, Java, SQL, TypeScript
Full Stack:	React, Node.js, Spring Boot, FastAPI, REST APIs, GraphQL, HTML, CSS, MUI
Database & Messaging:	MySQL, DynamoDB, MongoDB, Pub/Sub, RabbitMQ, ActiveMQ, Kafka
Cloud & DevOps:	AWS, GCP, Linux, Windows, Docker, CI/CD, Ansible, Kubernetes
Monitoring & Tools:	Postman, Swagger, Splunk, Pytest, Agile, Jira, SDLC, Jest, Cursor, Copilot, N8N, Salesforce
Certifications:	Salesforce Developer, Azure - 900, Nvidia Deep Learning

WORK EXPERIENCE

Intuit Credit Karma

May 2025 - Aug 2025

Software Engineer Intern

Charlotte, NC

- Developed and deployed an Experian mock report-pull **microservice** using **Scala**, integrating it into the gateway to enable test user support and save **7** days per outage on average by keeping org-wide testing running.
- Rolled out an **AI**-powered PR review pipeline across our team's services, using **LLM**-generated summaries and visualizations to cut PR review time 35% and speed merges.
- Built a data-quality alert using Scala and SQL in an Apache **Airflow** DAG, scanning **150K+** records for Experian PIN mismatches and sending Slack **alerts** to speed misroute fixes and avoid ~\$350K/year in penalties.
- Engineered **New Relic** dashboards for 5 services on a 120M+ platform, built an **ETL** to **BigQuery** for report-pull **failures**, and cut MTTD 80% and MTTR 40% via **on-call** alerting, strengthening releases and incident response.
- Migrated Credit Report publishing to **Google Pub/Sub** with WLID, removing SA keys for compliance and lower key overhead.

Vivance Pte. Ltd, Software Engineer

Dec 2023 - Jun 2024

- Devised core applications for the UV **medical** device from initial development using C, **Python**, and built simulation and debug tools using **JSON**, integrating with the company debug system to support unit and integration testing.
- Redesigned the UV device **firmware** architecture and migrated from STM32H7 to L4 in C, improving response time and throughput by 41%, cutting power use, and saving \$101 per device.
- Created a **React**-based visualization tool with a Python backend for **50GB+** datasets, replacing laggy Excel workflows with configurable plots and filters, cutting graph generation time by 43% for org-wide analysis.
- Implemented the **BLE652** upgrade and refactored firmware plus a boot-time Bash **auto-connect** script, improving device runtime by 32% and speeding connection setup across devices.

Goldman Sachs, Software Engineering Intern (Louisa AI, later spun out)

Feb 2023 - Jun 2023

- Automated networking-platform email delivery using **Java** and Amazon **SES**, building reusable **HTML** and **CSS** email templates to cut retries, manual ops, and **template** build time for 100K users, saving \$2.5K per year.
- Ingested news via **APIs** and ranked it using user preference signals to personalize daily digests, reducing manual curation and driving 1K+ user interactions per week.
- Debugged deployment issues via AWS **CloudWatch** and **Snowflake** logs, improving stability and cutting downtime by 40%.

PROJECTS

FFA-Lens: AI Retinal Lesion Detection | Elsevier Publication (9 Citations) – **RESEARCH PAPER**

Feb 2024

- Developed FFA-Lens using React, Python, PyTorch, **YOLO**, **Streamlit** to run lesion detection on FFA images, supporting single and **batch** uploads and exporting annotated outputs for clinician review and faster screening workflows.
- Trained on 5k labeled FFA images across 25 lesion classes, achieving **0.8+** precision and a **4.2** specialist rating on 500 test images.

AI Agent for Browser Task Automation

- Built a single-source **FastAPI** service where the user submits a prompt and an **AI agent**, powered by the Gemini API and **Playwright MCP**, interprets the webpage state to predict the next best action on the site.
- Containerized the application with **Docker** and implemented **CI/CD** using GitHub Actions to automate tests and deployment.

LEADERSHIP / EXTRACURRICULAR

- Vice President, ISTE Manipal**: Led the club during MIT flagship events, managing a team of **45–50**, driving recruitment, spearheading promotions for contests and initiatives, and overseeing **\$1,500** Tech Tatva budget.